

Theorem [Core] 6.1: Core $\equiv$ Matroid

$\mathfrak{F}$ is a Core FD Complex if and only if $\phi_{\mathfrak{F}}$ satisfies the Matroid Closure Axioms.

Proof: Since $\phi_{\mathfrak{F}}$ satisfy the axioms for closure operators, we need only show $\phi_{\mathfrak{F}}$ satisfies strong extension whenever $\phi_{\mathfrak{F}}$ satisfies (M-S EP).

(Strong Extension $\Rightarrow$ (M-S EP))

If a Core Complex $\mathfrak{F}$ satisfies (M-S EP) on its independent sets then $\mathfrak{F}$ satisfies (M-S EP), **Lemma [Core] 6.9.** $\mathfrak{F}$ is a Core Complex implies, $\mathfrak{F}$ satisfies (M-S EP) on independent sets, **Lemma [Core] 6.10.** Thus core FD-Complexes meet (M-S EP).

((M-S EP) $\Rightarrow$ Strong Extension)

This says that every Core FD Complex satisfies (M-S EP).

((M-S EP) $\Rightarrow$ Strong Extension)

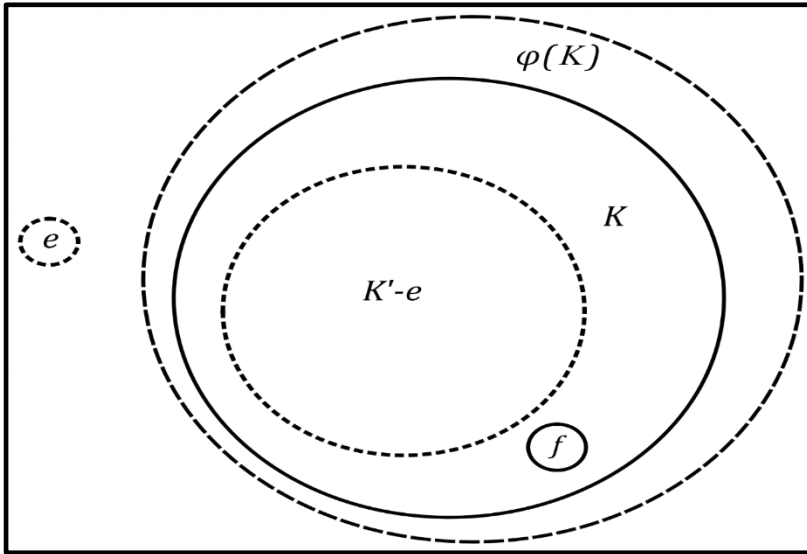


Figure [Core] 6.2: This Venn diagram illustrates how objects would be arranged if $\mathfrak{F}$ did not obey strong extension. This contradicts the proposition that $\mathfrak{F}$ satisfies (M-S EP). Consider the exterior of the diagram to be $\phi_{\mathfrak{F}}(K')$

Suppose $e \notin \phi_{\mathfrak{F}}(K)$ for a $K \in \mathcal{I}_{\mathfrak{F}}$ and $K \cup e \notin \mathcal{I}_{\mathfrak{F}}$. Weak extension gives a key $K' \subseteq K \cup e$ such that K' is a key of $\phi_{\mathfrak{F}}(K \cup e)$ and $e \in K'$. K' is a proper subset of $K \cup e$ because $K \cup e$ is dependent. There is an $f \in K \cup e$ such that $f \notin K'$ and $K' \rightarrow f \in \mathfrak{F}$. $f \neq e$ because $f \notin K'$. Now, $f \in \phi_{\mathfrak{F}}(K')$ because $f \in K \cup e \subseteq \phi_{\mathfrak{F}}(K')$ and $f \notin \phi_{\mathfrak{F}}(K' - e)$ because $f \subseteq K$, $f \notin K'$ and $\phi_{\mathfrak{F}}(K' - e) \cap K = K' - e \subseteq K'$. Note that if the last intersection includes any $g \in K$ that is not in $K' - e$ then